Optimization Homework 5 Dai Xunlian 225040015

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1 Q1

(a) **Convex.**

The set $S = \{s_1 + s_2 : s_1 \in S_1, s_2 \in S_2\}$ is the Minkowski sum of two convex sets S_1 and S_2. The Minkowski sum of convex sets is always convex. To prove this: for any $x, y \in S$, we have $x = s_1 + s_2$ and $y = t_1 + t_2$ where $s_1, t_1 \in S_1$ and $s_2, t_2 \in S_2$. Then for $\lambda \in [0, 1]$:

$$\lambda x + (1-\lambda)y = \lambda(s_1 + s_2) + (1-\lambda)(t_1 + t_2) = [\lambda s_1 + (1-\lambda)t_1] + [\lambda s_2 + (1-\lambda)t_2]$$

Since S_1 and S_2 are convex, the bracketed terms belong to S_1 and S_2 respectively, so $\lambda x + (1-\lambda)y \in S$.

(b) **Convex.**

The set $\{x \in \mathbb{R} : ax^2 + 2x + \frac{1}{a} \geq 0\}$ with $a > 0$ defines a sublevel set of a convex function. Since $a > 0$, the function $f(x) = ax^2 + 2x + \frac{1}{a}$ is convex (the second derivative $f''(x) = 2a > 0$). The sublevel set $\{x : f(x) \geq c\}$ of a convex function is convex.

(c) **Convex.**

The set $\{(y, x) \in \mathbb{R} \times \mathbb{R}^n : y \geq f(x)\}$ is the epigraph of the convex function f. The epigraph of any convex function is always a convex set. This follows from the definition of convexity: if f is convex, then for $(y_1, x_1), (y_2, x_2)$ in the epigraph and $\lambda \in [0, 1]$:

$$\lambda y_1 + (1-\lambda)y_2 \geq \lambda f(x_1) + (1-\lambda)f(x_2) \geq f(\lambda x_1 + (1-\lambda)x_2)$$

(d) **Not convex.**

The set $\{(y, x) \in \mathbb{R} \times \mathbb{R}^n : y = f(x)\}$ is the graph of the function f. The graph of a function is convex if and only if the function is affine (linear plus constant). Since f is only stated to be convex, not necessarily affine, the graph is generally not convex.

(e) **Not convex.** The set is defined by a quadratic inequality $x^\top Ax \leq 0$. The convexity of such a set depends on the eigenvalues of the symmetric part of A. Here, the symmetric matrix $B = \frac{1}{2}(A + A^\top)$ has eigenvalues of mixed signs (positive and negative), indicating that the set is not convex. Note that if x

were indeed in $\mathbb{R}^2$, the expression $x^\top Ax$ would be undefined due to dimension mismatch, so we assume $x \in \mathbb{R}^3$ for the analysis.

2 Q2

Are the following functions convex? Explain your answer.

(a) $f(x) = \|x\|_1 = \sum_{i=1}^d |x_i|$.

Answer: *Convex.* The absolute-value function $g(t) = |t|$ is convex on $\mathbb{R}$. A sum of convex functions is convex, hence $f(x) = \sum_{i=1}^d |x_i|$ is convex (equivalently, $\|\cdot\|_1$ is a norm and all norms are convex).

(b) $f : \mathbb{R}^d \to \mathbb{R}$ defined by $f(x) = \sum_{i=1}^d f_i(x_i)$ where each $f_i : \mathbb{R} \to \mathbb{R}$ is convex.

Answer: *Convex.* For any $x, y \in \mathbb{R}^d$ and $\lambda \in [0,1]$,

$$f(\lambda x+(1-\lambda)y) = \sum_{i=1}^d f_i(\lambda x_i+(1-\lambda)y_i) \leq \sum_{i=1}^d \big(\lambda f_i(x_i)+(1-\lambda)f_i(y_i)\big) = \lambda f(x)+(1-\lambda)f(y),$$

using convexity of each f_i.

(c) $f : \mathbb{R}^d \to \mathbb{R}$ satisfies $f(\lambda x) = |\lambda| f(x)$ for all $\lambda \in \mathbb{R}$ and $f(x+y) \leq f(x) + f(y)$ for all x, y.

Answer: *Convex.* For $t \in [0,1]$,

$$f(tx + (1-t)y) \leq f(tx) + f((1-t)y) = tf(x) + (1-t)f(y),$$

where we used subadditivity and absolute homogeneity. Thus f satisfies the convexity inequality.

(d) $f(x) = |x|^{3/2}$ on $\mathbb{R}$.

Answer: *Convex.* For $x > 0$ one has $f(x) = x^{3/2}$, $f'(x) = \frac{3}{2}x^{1/2}$ and $f''(x) = \frac{3}{4}x^{-1/2} > 0$. For $x < 0$ the function is even ($f(-x) = f(x)$) and the second derivative on $(-\infty, 0)$ has the same positive expression in $|x|$. Hence $f''(x) \geq 0$ for all $x \neq 0$ and the function is convex (the behavior at 0 is compatible with convexity).

(e) $f(x) = \ln\left(x + \sqrt{1+x^2}\right) + x^2$.

Answer: *Convex.* Note $\ln\left(x + \sqrt{1+x^2}\right) = \operatorname{arsinh}(x)$ and

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} \operatorname{arsinh}(x) = -\frac{x}{(1+x^2)^{3/2}}.$$

Therefore

$$f''(x) = 2 - \frac{x}{(1+x^2)^{3/2}}.$$

The function $h(x) = \dfrac{x}{(1+x^2)^{3/2}}$ attains its maximum value ≈ 0.3849 at $x = 1/\sqrt{2}$, so

$$f''(x) \geq 2 - 0.3849 > 0 \quad \text{for all } x,$$

hence $f''(x) > 0$ everywhere and f is strictly convex.

3 Q3

Suppose we have centered observations $\{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$ and

$$\sum_{i=1}^n x_i = 0, \qquad \sum_{i=1}^n y_i = 0.$$

Let $(\beta_0^\star, \beta^\star)$ be a global minimizer of

$$f(\beta_0, \beta) = \sum_{i=1}^n \big(\beta_0 + \beta^\top x_i - y_i\big)^2, \qquad \beta_0 \in \mathbb{R},\ \beta \in \mathbb{R}^d.$$

(a) **Show that** $\beta_0^\star = 0$.

Proof. Differentiate f w.r.t. β_0 and set to zero (first-order condition for a minimizer):

$$\frac{\partial f}{\partial \beta_0} = 2\sum_{i=1}^n \big(\beta_0 + \beta^\top x_i - y_i\big) = 0.$$

Hence

$$n\beta_0 + \sum_{i=1}^n \beta^\top x_i - \sum_{i=1}^n y_i = 0.$$

Using $\sum_i x_i = 0$ and $\sum_i y_i = 0$ gives $n\beta_0 = 0$, so $\beta_0 = 0$. Thus any global minimizer satisfies $\beta_0^\star = 0$.

(b) The observations before centering are $x_i' = x_i + q$, $y_i' = y_i + r$ with $q \in \mathbb{R}^d$, $r \in \mathbb{R}$. Show that (β_0, β) minimizes f iff $(\beta_0 - \beta^\top q + r, \beta)$ minimizes

$$\tilde{f}(\beta_0, \beta) = \sum_{i=1}^n \big(\beta_0 + \beta^\top x_i' - y_i'\big)^2.$$

Proof. Substitute x_i', y_i':

$$\beta_0 + \beta^\top x_i' - y_i' = \beta_0 + \beta^\top (x_i + q) - (y_i + r) = (\beta_0 + \beta^\top q - r) + \beta^\top x_i - y_i.$$

Therefore

$$\tilde{f}(\beta_0, \beta) = \sum_{i=1}^{n} \big((\beta_0 + \beta^\top q - r) + \beta^\top x_i - y_i\big)^2 = f\big(\beta_0 + \beta^\top q - r,\ \beta\big).$$

Hence minimizing $\tilde{f}(\beta_0, \beta)$ over (β_0, β) is equivalent to minimizing $f(\tilde{\beta}_0, \beta)$ over $(\tilde{\beta}_0, \beta)$ with the relation $\tilde{\beta}_0 = \beta_0 + \beta^\top q - r$. Equivalently, (β_0, β) minimizes f iff $(\beta_0 - \beta^\top q + r, \beta)$ minimizes $\tilde{f}$.

4 Q4

5 Q4

(a) Convexity. The objective function is

$$f(x) = 3x_1^2 + 4x_2^2 + 2x_1x_2,$$

whose Hessian is

$$\nabla^2 f = \begin{pmatrix} 6 & 2 \\ 2 & 8 \end{pmatrix}.$$

Since the Hessian is positive definite (determinant $44 > 0$ and trace $14 > 0$), the objective is strictly convex.

The constraint function is

$$g(x) = \sqrt{1 + x_1^2 + x_2^2} + x_1 - 2.$$

The function $x \mapsto \sqrt{1 + \|x\|^2}$ is convex, and adding a linear term preserves convexity; hence g is convex. For a convex optimization problem, the constraint $g(x) \le 0$ requires g to be convex. Therefore, the feasible set is convex and the problem is a convex optimization problem.

(b) KKT conditions. Let $\lambda \ge 0$ be the Lagrange multiplier. The Lagrangian is

$$\mathcal{L}(x, \lambda) = f(x) + \lambda\, g(x).$$

The gradients are

$$\nabla f(x) = \begin{pmatrix} 6x_1 + 2x_2 \\ 2x_1 + 8x_2 \end{pmatrix}, \qquad \nabla g(x) = \begin{pmatrix} \dfrac{x_1}{\sqrt{1 + x_1^2 + x_2^2}} + 1 \\ \dfrac{x_2}{\sqrt{1 + x_1^2 + x_2^2}} \end{pmatrix}.$$

The KKT conditions are:

$$\begin{aligned}
&\text{(stationarity)} \qquad \nabla f(x) + \lambda\, \nabla g(x) = 0, \\
&\text{(primal feasibility)} \qquad g(x) \le 0, \\
&\text{(dual feasibility)} \qquad \lambda \ge 0, \\
&\text{(complementary slackness)} \qquad \lambda\, g(x) = 0.
\end{aligned}$$

(c) KKT points.
Case 1: $\lambda = 0$. The stationarity condition reduces to $\nabla f(x) = 0$:

$$\begin{cases} 6x_1 + 2x_2 = 0, \\ 2x_1 + 8x_2 = 0. \end{cases}$$

Solving yields the unique solution $x = (0, 0)$. Primal feasibility: $g(0, 0) = 1-2 = -1 \leq 0$. Hence $(x, \lambda) = ((0, 0), 0)$ is a KKT point.

Case 2: $\lambda > 0$. Complementary slackness requires $g(x) = 0$, i.e.,

$$\sqrt{1 + x_1^2 + x_2^2} + x_1 - 2 = 0.$$

Together with the stationarity equations, numerical solving shows that the resulting Lagrange multiplier is negative, violating $\lambda \geq 0$. Thus no KKT point exists with $\lambda > 0$.

Therefore, the only KKT point of the problem is

$$\boxed{x^* = (0, 0), \qquad \lambda^* = 0.}$$

6 Q5

Consider the optimization problem

$$\min_{x \in [0,1]} e^x - 2x.$$

We solve it using both the Bisection Method and the Golden Section Method, stopping when the interval length becomes smaller than 10^{-4} and taking the midpoint of the final interval as the approximate minimizer.

(a) Bisection Method

To apply the Bisection Method, we solve for the stationary point by finding the root of the derivative

$$f'(x) = e^x - 2.$$

Since $f'(0) = -1 < 0$ and $f'(1) = e - 2 > 0$, there is a root in $[0, 1]$. Running the method with tolerance 10^{-4} yields:

$$x^\star_{\text{bisect}} \approx 0.693145751953125, \qquad \text{iterations} = 14.$$

This result is close to the analytical minimizer $\ln 2 \approx 0.693147$.

(b) Golden Section Method

We now minimize the original objective function using the Golden Section search over $[0, 1]$. With the same tolerance 10^{-4}, the method gives:

$$x^\star_{\text{golden}} \approx 0.6931367391457772, \qquad \text{iterations} = 20.$$

(c) Summary

Both numerical methods return values consistent with the true minimizer $x^\star = \ln 2$. The Bisection Method converges faster in this case because it operates on the strictly monotone derivative $f'(x)$.